Ground Subsidence Observations

Modified Influence Function Method for Complete Subsidence Prediction

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Paper focus

Indian coalfields, irregular room-and-pillar panels and complete subsidence prediction.

Core contribution

Influence function + edge, asymmetry and overlying-goaf corrections.

Presentation

Technical study with key findings, methodology and Indian coalfield case studies.

1. Problem statement and research motivation

Indian underground coal extraction is dominated by room-and-pillar mining, producing panels with irregular shapes.

Earlier empirical approaches were largely two-dimensional and implicitly assumed plane-strain geometry.

Most Indian coal mining is at shallow to moderate depth (<350 m), where discontinuous/stepped subsidence is observed.

UK Subsidence Engineers' Handbook norms were found to be inapplicable to Indian geomining conditions.

Extraction below previously caved goaf produces different—and generally larger—subsidence than extraction below unfractured strata.

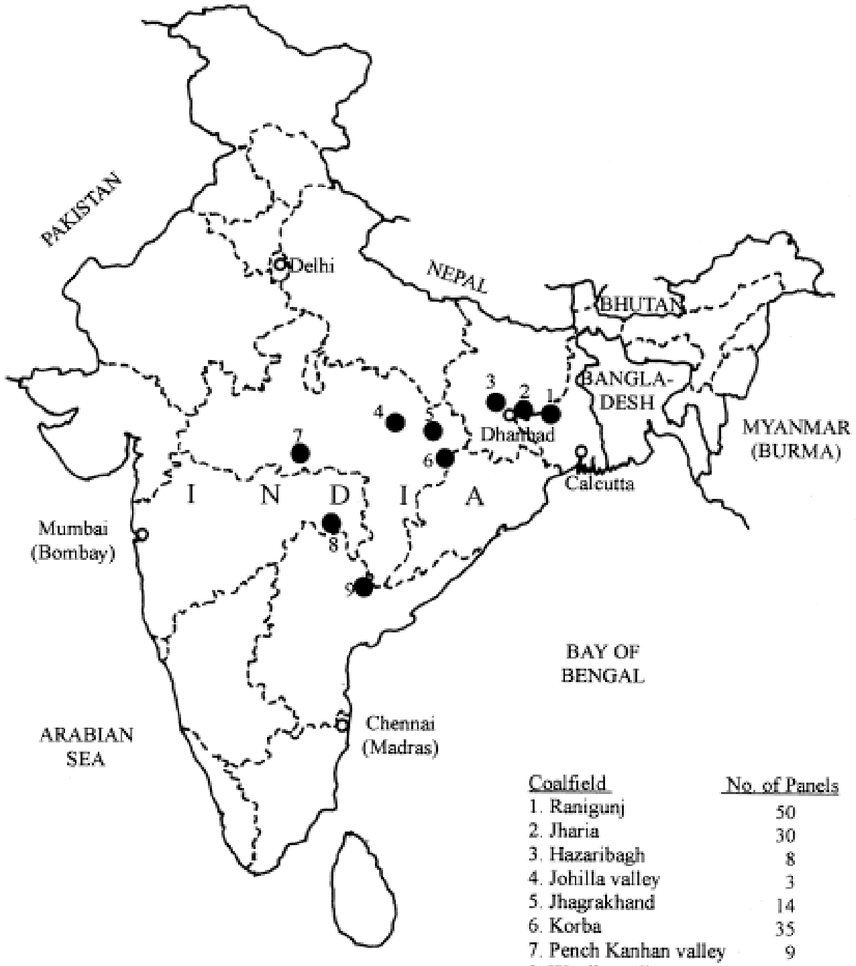
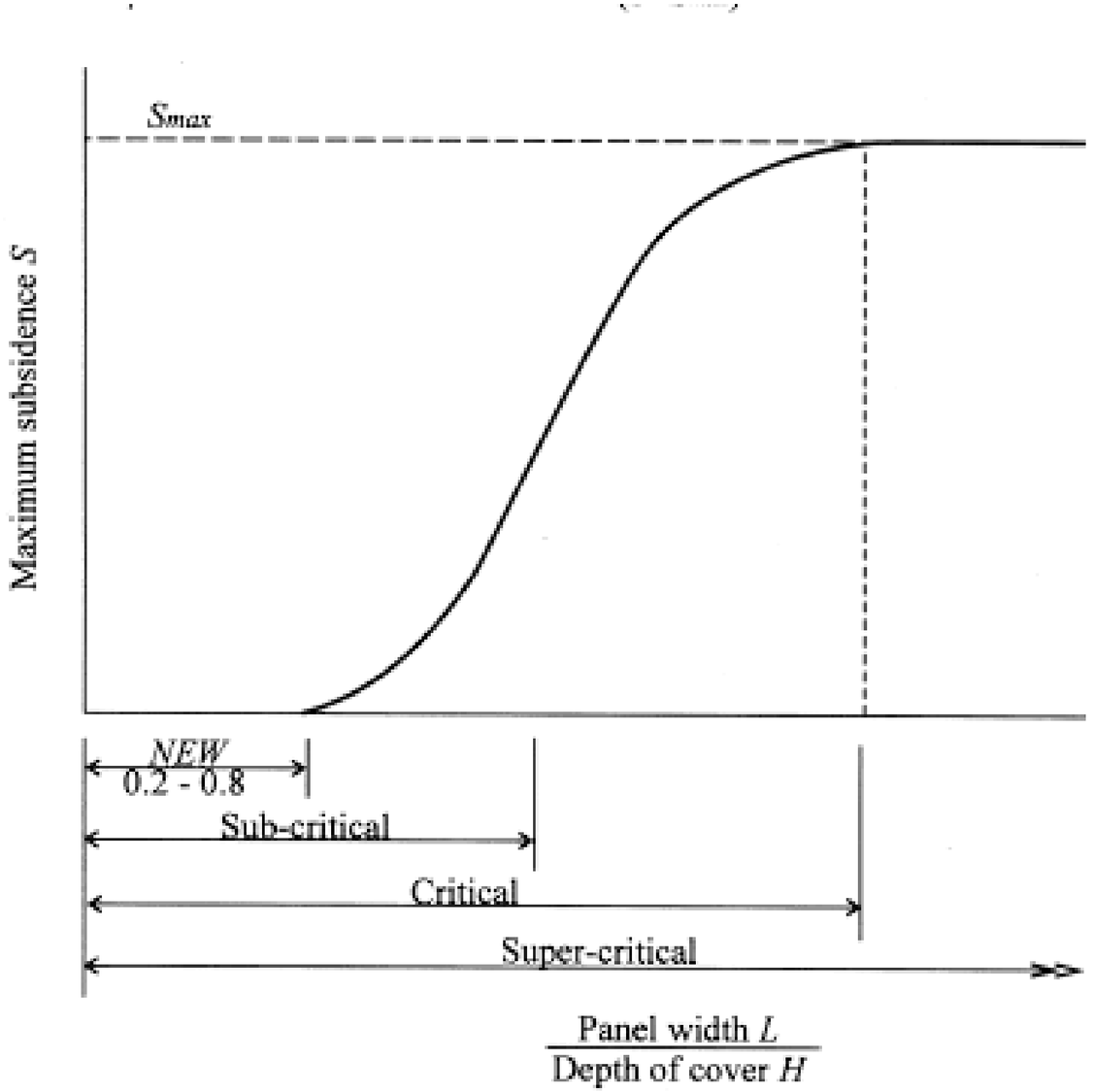


Fig. 2. Map of India showing the coalfields where subsidence studies were conducted.

2. Basic subsidence concepts



NEW
Non-effective width-to-depth ratio: maximum extraction width with no significant surface symptom.

Critical width

Panel width at which maximum subsidence is approached.

$$x = L / (H \times NEW)$$

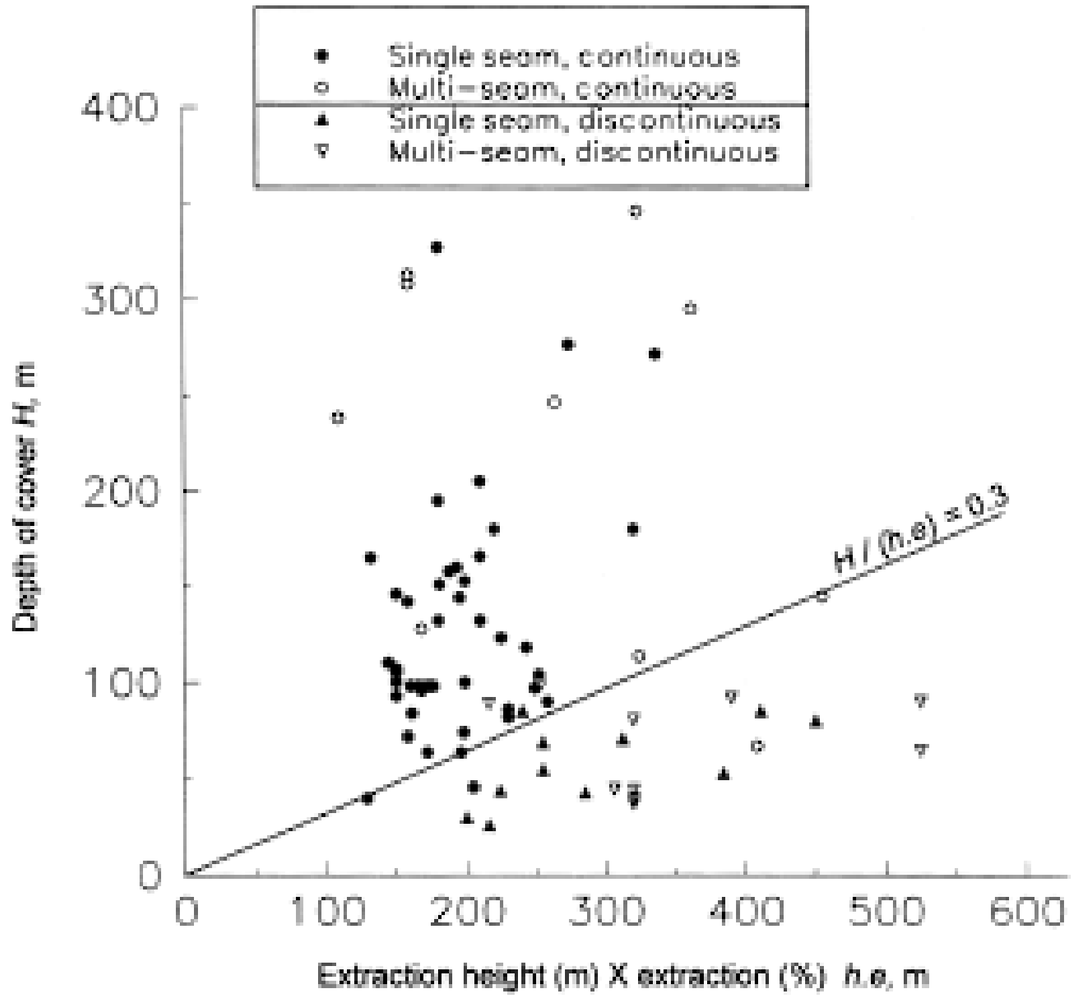
$$s = Smax / (h \cdot e)$$

Angle of draw γ

Lateral extent of mining influence; the paper reports substantial variation with strata conditions.

3. Discontinuous subsidence — the safety limit

One of the most important observations for shallow Indian workings



H / (h·e) < 0.3 → discontinuous subsidence

Subsidence occurs in shear steps rather than as a smooth continuous trough.
Surface damage from discontinuous subsidence is particularly severe.
The demarcation line was developed from observed Indian cases and applies to both single- and multi-seam conditions.
Pot-holing/chimneying is classified as discontinuous subsidence but is not treated in this paper.

4. Maximum subsidence — single vs multi-seam

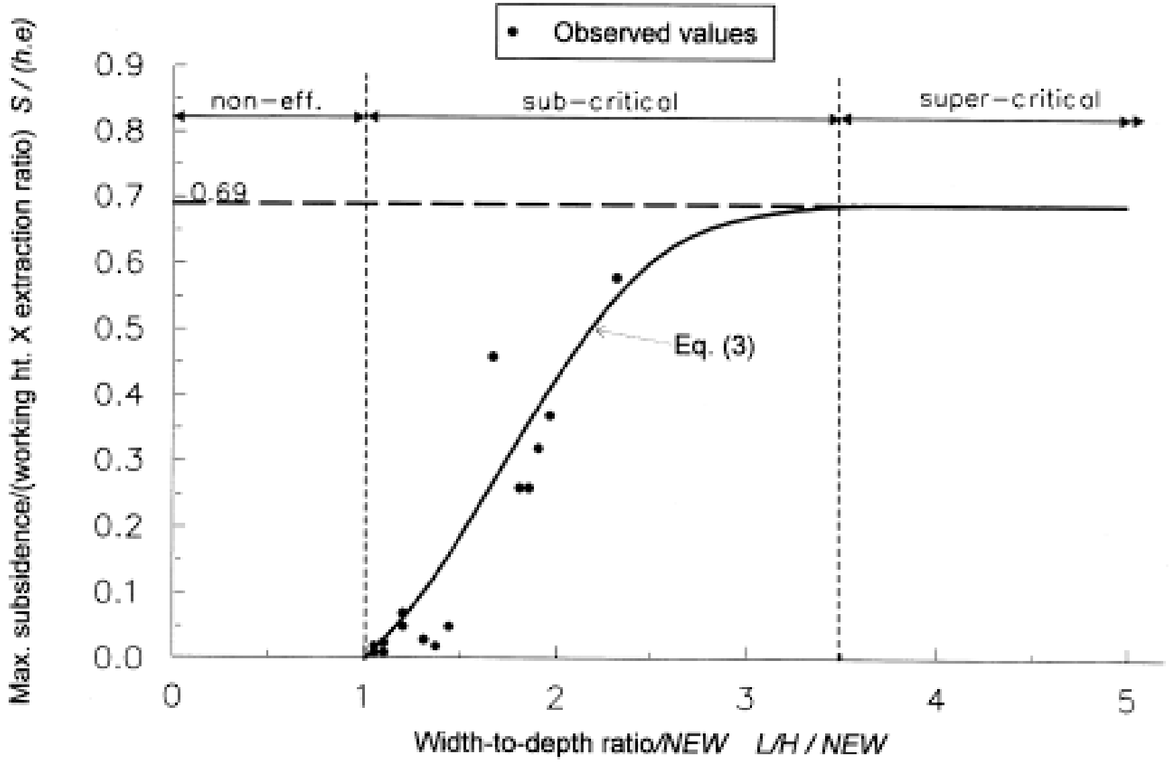


Fig. 4. Maximum subsidence for single seam cases.

Single seam: s ≈ 0.69

Multi-seam: s ≈ 0.81

5. Empirical subsidence equations and width regimes

Single seam: $S = 0.33[1 + 1.1 \tanh(1.4(x - 1.8))]$

Multi-seam: $S = 0.12[1 + 5.9 \tanh(0.38(x - 1.45))]$

Non-effective

$x < 1$
Little/no significant surface symptom.

Sub-critical

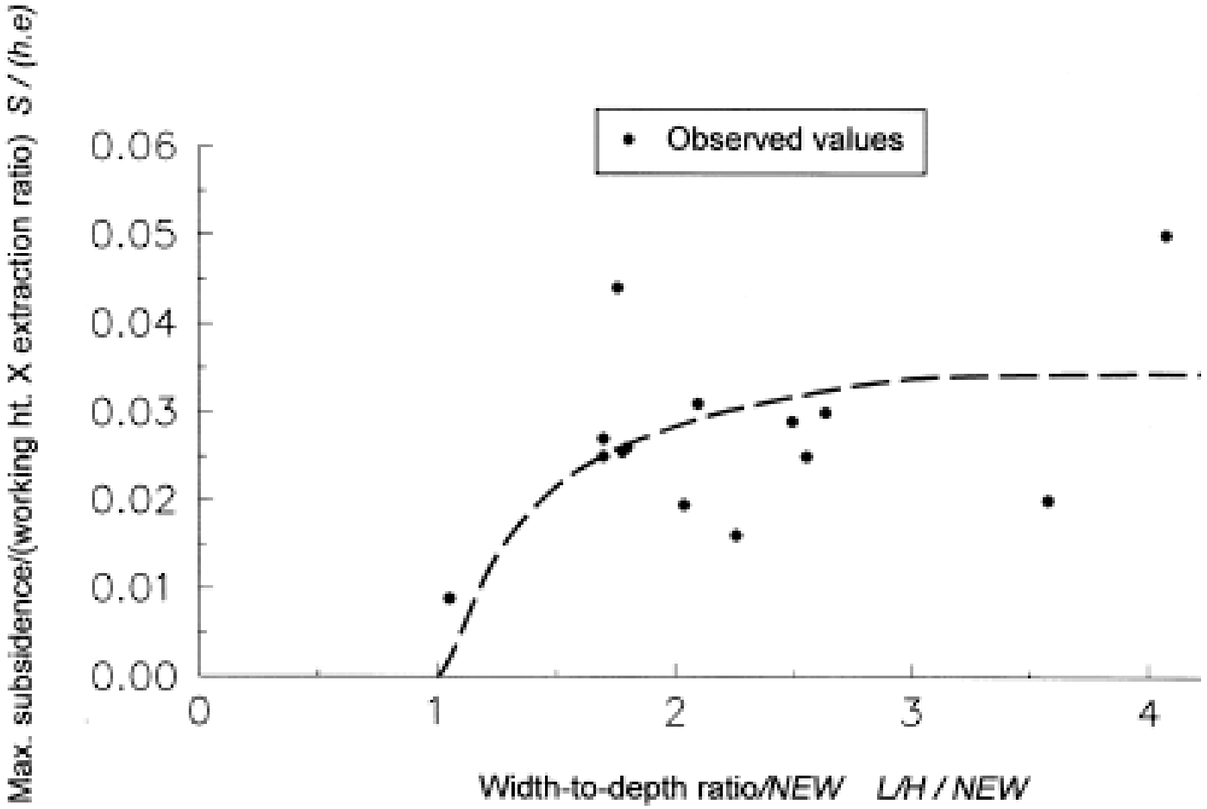
$1<x<3.5$
Subsidence rises rapidly as effective width grows.

Super-critical

$x>3.5$
Beyond critical width, subsidence approaches Smax.

Fit quality: $R^2 = 0.85$ for single seam and 0.86 for multi-seam.
Critical width ≈ 3.5×NEW for single seam; ≈ 6.5–7×NEW for multi-seam.
Multi-seam Eq. (4) is only approximate because overlying goaf geometry can be irregular.

6. Stowing and angle of draw



Hydraulic stowing

For good stowing, the paper proposes an approximate subsidence factor of 0.05. Poor stowing makes prediction unreliable.

Angle of draw

Observed Indian values vary roughly from 9° to 25°, depending on strata conditions.

Draw should ideally be measured normal to the panel side using a long rectangular panel.
Panel corners significantly affect the draw limit.
A panel needs sufficient width and length for the draw to develop around the whole panel.
Panels only slightly wider than NEW may show negative draw at shorter sides and positive draw at longer sides.

7. Maximum slope and strain

$$G = K_1 S/H$$

$$E(-) = K_2 S/H$$

$$E(+) = K_3 S/H$$

G — maximum slope

Measures how rapidly subsidence changes horizontally.

E(−) — compression

Maximum compressive strain.

E(+) — tension

Maximum tensile strain; relevant to cracking/distortion.

8. Conventional influence-function method

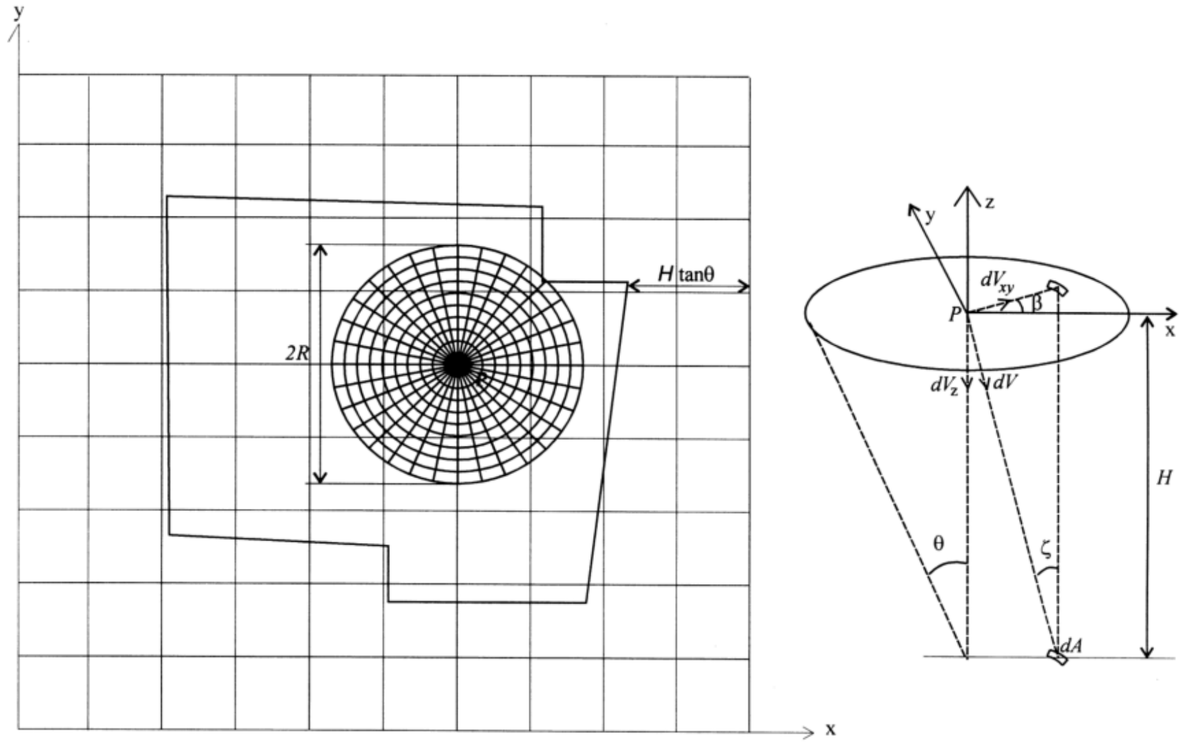


Fig. 10. Illustration of subsidence and horizontal displacements calculation using influence function method.

Lay a surface grid over the extraction panel and extend it to the draw limits.

At each grid point P, construct a circular influence area with radius R = H tanγ.

Divide the circle into rings and sectors; the paper uses 10 rings × 64 sectors = 640 elements.

Only sectors/elements falling inside the extracted area contribute to subsidence.

Sum the weighting factors and multiply by Smax.

$$\mathrm{d}V_{xy} = \tan\ \zeta \mathrm{d}V_{\mathrm{z}} \quad (\mathrm{d}V_{\mathrm{z}} = S_{ie} S_{\max})$$

9. Modified influence function

$$Kz = 0.5352/R^2 \cdot [1 + \cos(\pi r/R)]$$

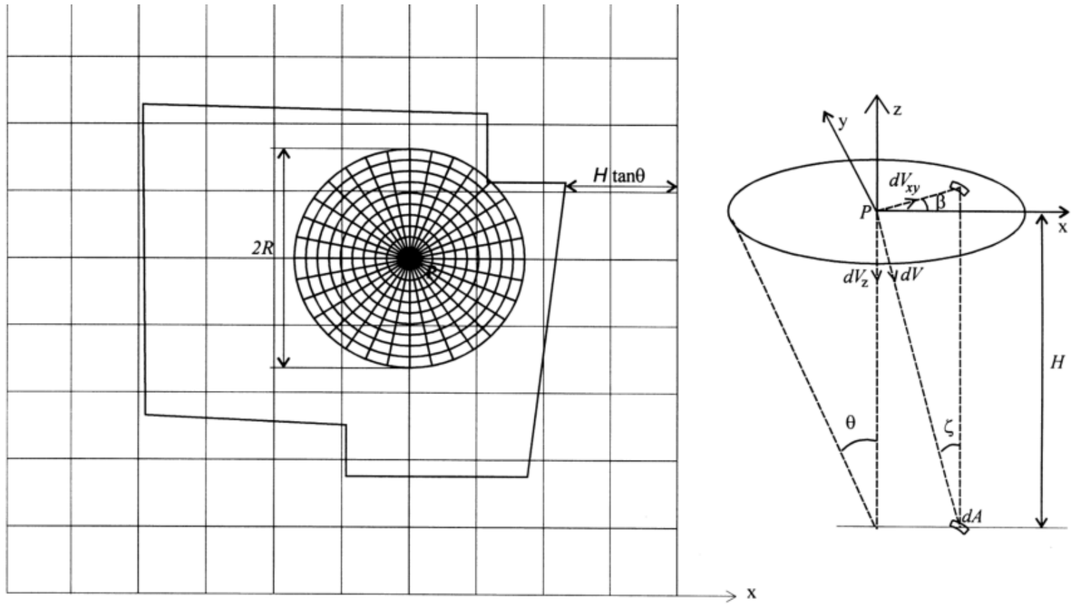


Fig. 10. Illustration of subsidence and horizontal displacements calculation using influence function method.

Why modify Kz?

The choice is empirical, but the influence function must satisfy the normalization condition ∫∫ Kz dA = 1.

Ring weighting factors

For 10 rings the paper gives weights whose sum is exactly 1.0.

Important

The new function is the base on which the edge and asymmetry corrections are applied.

10. Edge and asymmetry corrections

$$W_z = 0.5 \tanh[5db/(1.5\ NEW \cdot H) - 2.4] + 0.5$$

$$Q_z = 0.9 - 0.1 \tanh[0.5(d - 0.4d_{max})]$$

$$dV'_z = Q_z \cdot W_z \cdot dV_z$$

Edge correction — Wz

The conventional method tends to predict excessive subsidence near panel edges. db is distance to the nearest extraction boundary.

Asymmetry correction — Qz

Indian subsidence troughs are asymmetric; maximum subsidence occurs closer to the panel start line.

11. Effect of overlying goaf in multi-seam extraction

$$NEW' = [0.6\ NEW(H - T) + NEW(T)] / H$$

$$a' = [1.17\ a(H - T) + a(T)] / H$$

40% NEW reduction

Within elements falling inside the overlying goaf, the original NEW is reduced by 40%.

17% factor increase

The subsidence factor is increased by 17% for goaf-influenced elements.

Weighted average

The intact rock parting remains relevant, so the final evaluation uses weighted averaging.

The paper reports the average factor rising from 0.69 (single seam) to 0.81 (multi-seam), about 17%.

Only elements within the overlying goaf receive the goaf-specific changes.

The method was adopted after trials across cases with different goaf shapes and parting thicknesses.

12. Computer implementation and validation

Input: depth H, NEW, angles, extraction geometry, grid spacing and overlying-goaf geometry.

Generate surface grid points and initialize displacement arrays.

For every surface point, construct the influence circle and divide it into sectors/elements.

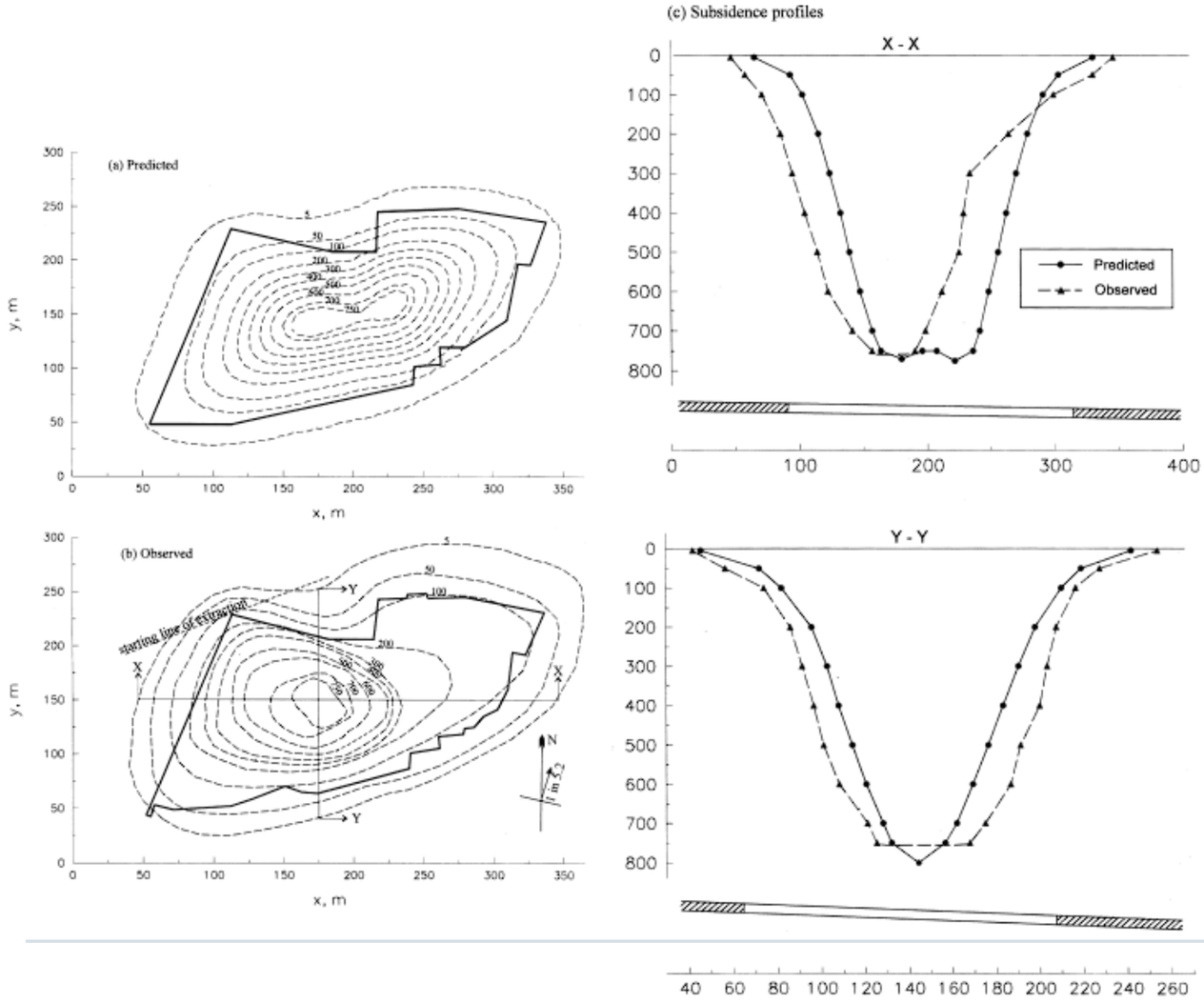
Evaluate weighting, apply goaf changes where required, accumulate vertical and horizontal displacement.
Finally calculate slopes and strains and print the predicted outputs.

FLOWCHART

Start → Input parameters → Determine grid → Generate surface points → Initialize arrays → Construct influence circle → Divide into rings & sectors → Check extracted area → Calculate weighting factor → Check overlying goaf → Modify aa, NEW if required → Calculate subsidence → Calculate horizontal displacement → Check elements → Calculate slope & strain → Check surface points → Print/plot results → End

8 single-seam cases + 5 multi-seam cases were tested; predicted and observed values were reported as satisfactory.

13. Case study — Sukri Colliery



Mine data

Average depth = 165 m
Seam dip ≈ 1 in 5.2
Extraction height = 2.4 m
Extraction = 75%

Field observations

NEW = 0.45
Angle of draw ≈ 13°

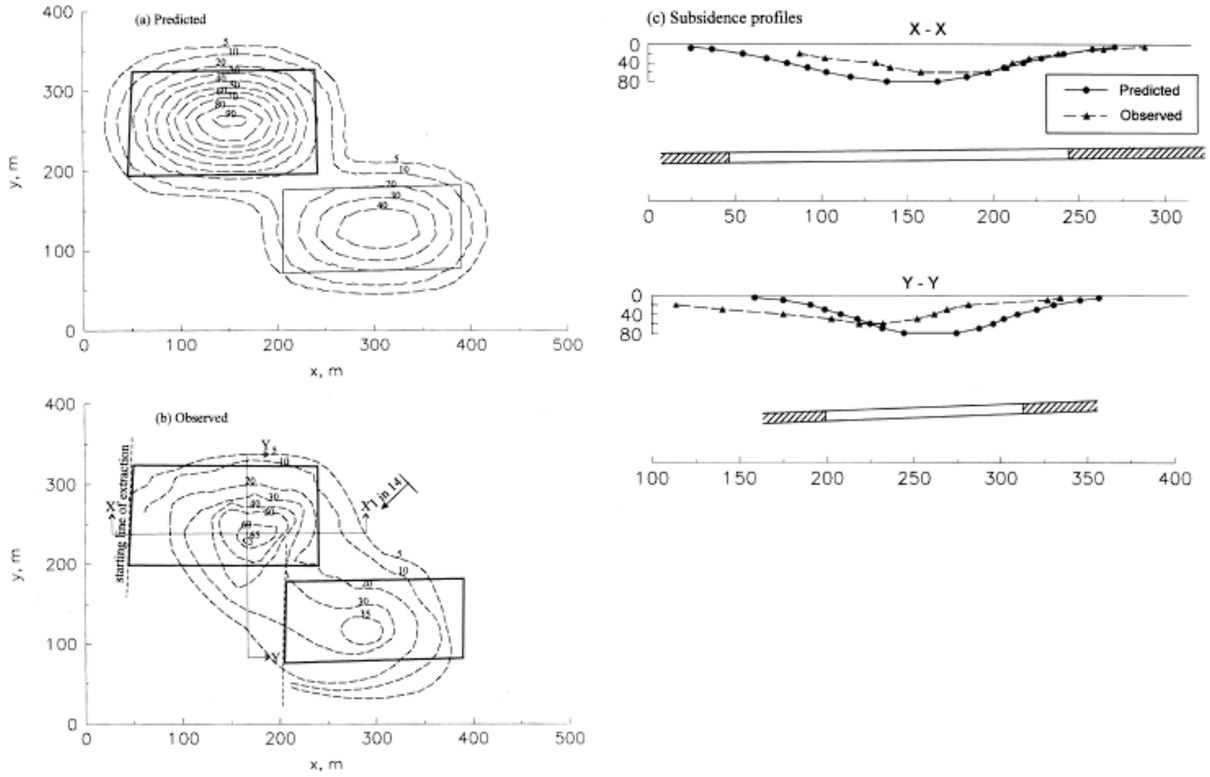
What the figure shows

Predicted and observed contours are compared on the panel plan, with x–x and y–y subsidence profiles.

Limitation exposed

Agreement is less strong when seam inclination begins to influence subsidence.

14. Case study — Ningah Colliery



Mine data

Average depth = 180 m
Extraction height = 2.0 m
Extraction = 100%

Field observations

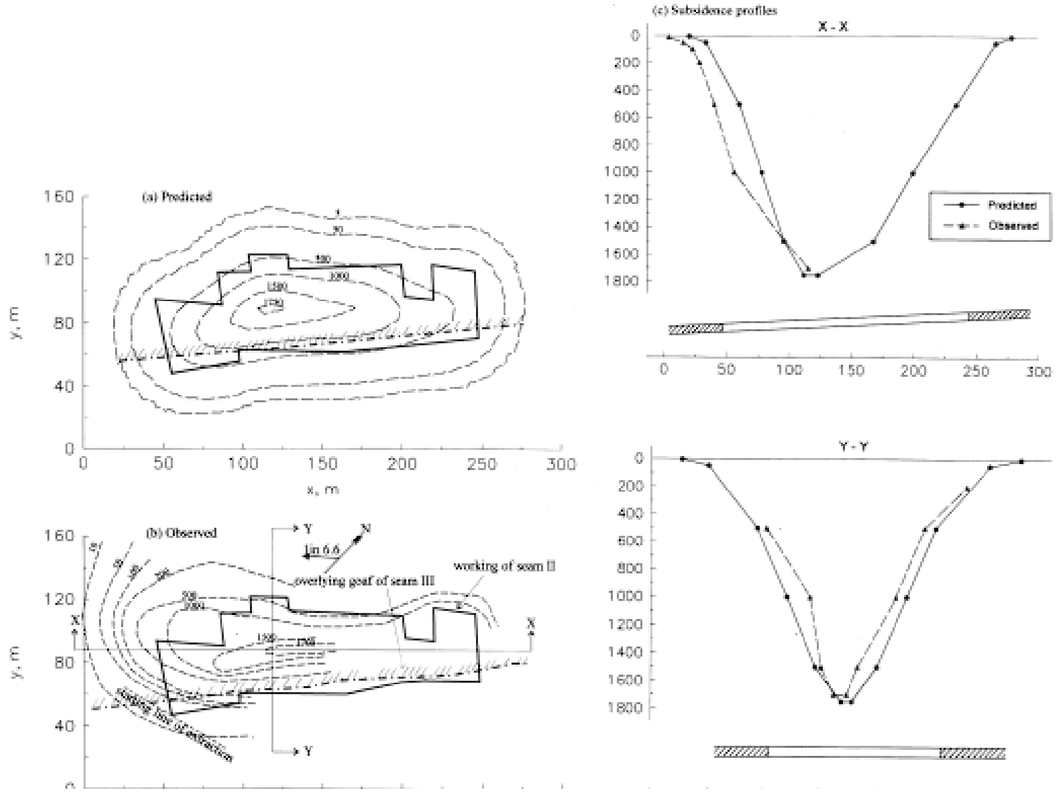
NEW = 0.7
Angle of draw ≈ 20°

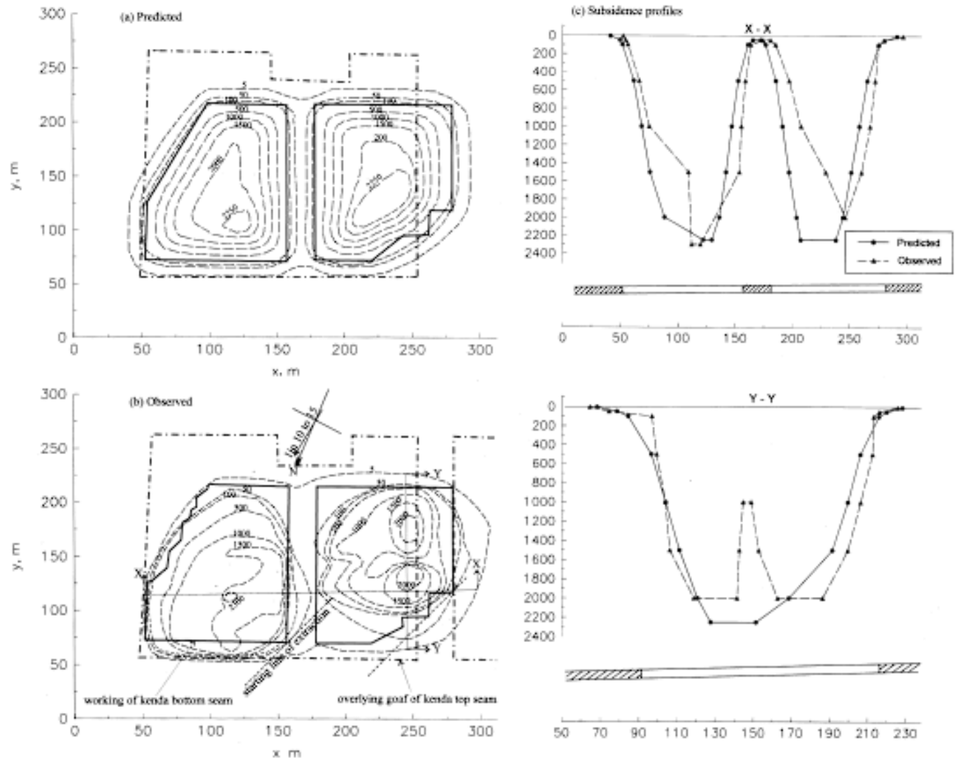
Panel widths were close to the observed non-effective width.

Predicted and observed contours are compared with x–x and y–y subsidence profiles.

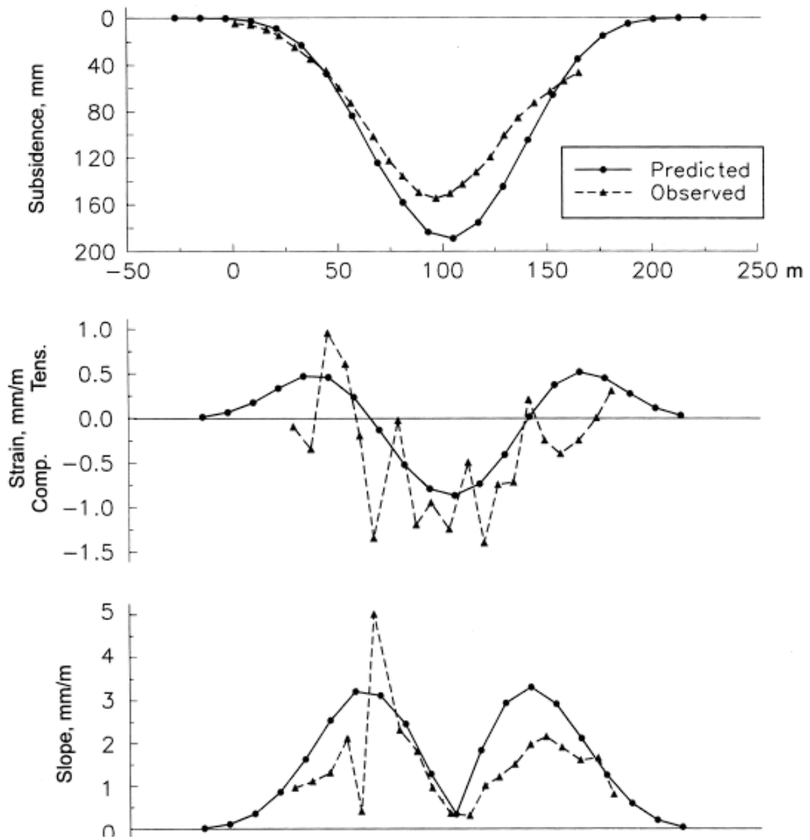
This provides a useful longwall comparison with the irregular room-and-pillar Sukri case.

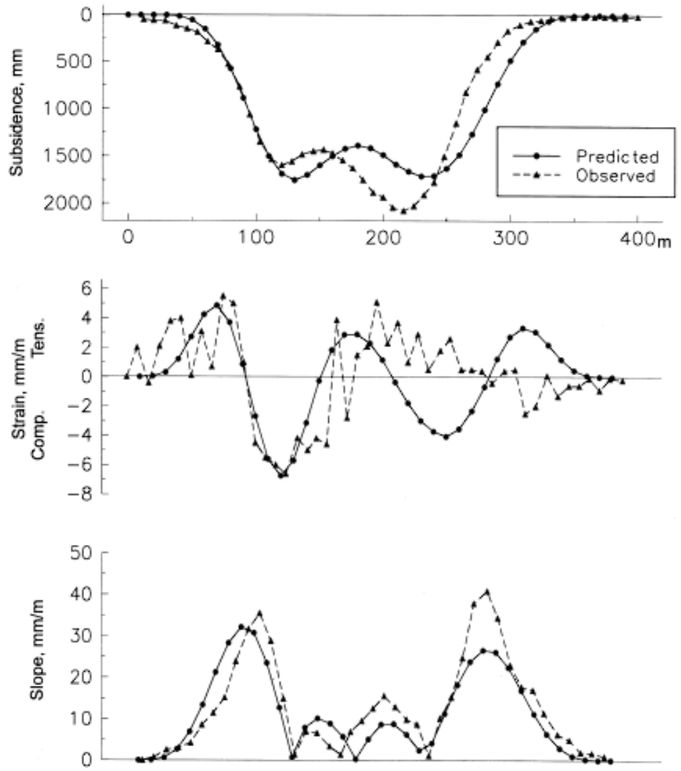
15. Multi-seam case studies — Bera & North Searsole





16. Slope/strain validation + final conclusions





The modified influence function is reported as reasonable for different panel shapes and sizes after correcting asymmetry, edge effects and overlying goaf. The method is intended for horizontal or near-horizontal seams; inclined seams can show poorer agreement.

Extraction below goaf increases subsidence factor, slope and strain.

Some unusually low central-Indian subsidence cases remain unexplained; the paper points to observation duration, in-situ stress and bulking as possible factors.

CONCLUSION

The study presents a modified influence function approach for predicting ground subsidence associated with underground coal mining. By incorporating the effects of panel geometry, edge influence, subsidence asymmetry, and overlying goaf, the method provides a more realistic representation of observed ground movement in Indian coalfields.

The analysis and case studies demonstrate the applicability of the approach for evaluating subsidence, slope, and strain, thereby providing useful support for assessing potential surface impacts and improving underground mine planning

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